

Incident Lifecycle Automation in ServiceNow

1. Ideation Phase

1.1 Project Title

Incident Lifecycle Automation in ServiceNow

1.2 Project Overview

The project aims to automate and standardize the complete **Incident Management lifecycle** using ServiceNow. The proposed solution enables Service Desk agents to create, classify, investigate, escalate, resolve, and track incidents while integrating Knowledge Management, Change Management, SLA tracking, and Child Incident Management.

The idea is to establish a structured incident-handling process where an incident can move seamlessly from initial reporting to final resolution while maintaining proper documentation and traceability.

1.3 Problem Statement

Organizations frequently receive IT-related issues such as network failures, VPN authentication problems, hardware issues, and application failures. When these incidents are handled manually or inconsistently, several problems may occur:

- Incorrect or incomplete incident classification.
- Delayed assignment to the appropriate support team.
- Difficulty finding existing solutions.
- Poor visibility into incident progress.
- Lack of proper escalation mechanisms.
- Manual creation and tracking of related change requests.
- Difficulty tracking recurring or related incidents.
- Incomplete cause and resolution documentation.
- Poor SLA compliance.

Therefore, there is a need for an integrated and standardized incident management solution.

1.4 Proposed Idea

The proposed ServiceNow solution automates the incident lifecycle by providing:

- Incident creation through Service Operations Workspace.
- Standardized incident classification.
- Knowledge article recommendations through Agent Assist.
- Assignment and escalation to appropriate support groups.
- Emergency Change Request integration.
- Child Incident creation.
- Cause and resolution documentation.
- Knowledge Article creation from resolved incidents.
- SLA and related-record tracking.
- Multi-team collaboration and incident visibility.

1.5 Project Goals

The major goals of the project are:

1. Standardize incident creation and classification.
2. Reduce incident resolution time.
3. Improve incident assignment and escalation.
4. Reuse existing knowledge for faster troubleshooting.
5. Integrate Incident and Change Management.
6. Track recurring issues using child incidents.
7. Maintain proper cause and resolution information.
8. Convert useful incident resolutions into Knowledge Articles.
9. Improve SLA compliance and service visibility.
10. Provide an end-to-end incident lifecycle.

2. Requirement Analysis

2.1 Business Requirements

The system should support the following business requirements:

Requirement	Description
Incident Creation	Service Desk agents should be able to create incidents.
Classification	Incidents should contain appropriate category, subcategory, urgency, service, and CI information.
Knowledge Integration	Agents should be able to search and attach relevant Knowledge Articles.
Assignment	Incidents should be assigned to appropriate support groups.
Escalation	Level 2 teams should be able to investigate and resolve incidents.
Change Integration	Emergency changes should be associated with incidents.
Child Incidents	Related or recurring issues should be tracked through child incidents.
Resolution	Cause, resolution code, and resolution notes should be documented.
Knowledge Creation	Resolved incidents should be converted into Knowledge Articles where appropriate.
SLA Tracking	SLA information should be visible throughout the incident lifecycle.

2.2 Functional Requirements

FR-01: Service Creation

The system should support creation of a service:

- **Service:** Remote Access
- **Operational Status:** Operational

FR-02: Service Offering Creation

A service offering should be created under the Remote Access service:

- **Service Offering:** Corporate VPN
- **Parent Service:** Remote Access
- **Operational Status:** Operational

FR-03: Incident Creation

The Service Desk agent should be able to create an incident with:

- Short Description
- Caller
- Assignment Group
- Description
- Channel
- Category
- Subcategory
- Urgency
- Service
- Service Offering
- Configuration Item

FR-04: Incident Classification

The incident should be classified appropriately.

Example:

- **Category:** Network
- **Subcategory:** VPN
- **Urgency:** 2 – Medium
- **Service:** Remote Access
- **Service Offering:** Corporate VPN
- **Configuration Item:** ThinkStationS20

FR-05: Knowledge Integration

Agent Assist should recommend relevant Knowledge Articles based on the incident description.

The agent should be able to:

- View recommended articles.
- Open an article.
- Mark an article as Helpful.
- Attach the article to the incident.
- Add comments regarding the article.

FR-06: Watch List and Work Notes

The system should allow users to be added to:

- Watch List
- Work Notes List

This allows stakeholders to monitor incident progress and updates.

FR-07: Reassignment

The incident should be reassigned from the Service Desk to the appropriate Level 2 support group.

Example:

Service Desk → Network

FR-08: Investigation

The Level 2 Network team should be able to:

- View incident details.
- View Knowledge Articles.
- Review activity.
- View SLA information.
- Take ownership of the incident.

FR-09: Change Integration

When a change is required to resolve the incident, the incident should support creation and tracking of an emergency change request.

The incident can be placed in:

On Hold → Awaiting Change

FR-10: Child Incident

The system should allow creation of child incidents for recurring or related issues.

The child incident should be linked to the parent incident.

FR-11: Cause Documentation

The probable cause of the incident should be documented.

Example:

PowerEdge service was suspended and required restart.

FR-12: Resolution Documentation

The system should record:

- Resolution Code
- Resolution Notes

Example:

- **Resolution Code:** Workaround provided
- **Resolution Notes:** Restarted VPN-SRV-02 service as per emergency change request.

FR-13: Knowledge Article Creation

After resolving the incident, the solution should be reusable by creating a Knowledge Article.

Requirements:

- Knowledge Base: IT
- Template: Standard
- Article should contain relevant incident resolution information.

FR-14: SLA Tracking

The system should provide visibility into applicable SLAs through the incident's related records.

3. Project Design Phase

3.1 Solution Architecture

The proposed solution uses ServiceNow's integrated modules to manage the complete incident lifecycle.

High-Level Flow:



3.2 Service Structure

The service hierarchy is designed as:

Remote Access

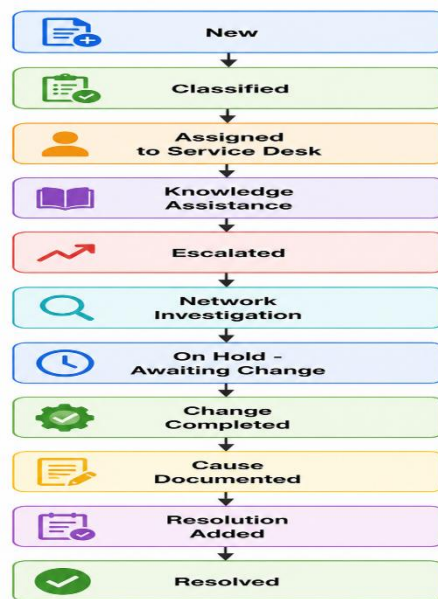


Where:

- **Remote Access** = Parent Service
- **Corporate VPN** = Service Offering

3.3 Incident Lifecycle Design

The incident lifecycle is designed as:



3.4 Stakeholder Design

Stakeholder	Role	Expectations	Impact of Automation
End Users	Report IT issues	Quick resolution and visibility	Faster resolution and transparent tracking
Service Desk Agents	First-level support	Easy logging and classification	Standardized intake and faster escalation
Level 2 Support	Diagnose and resolve issues	Complete incident information	Better diagnosis and clear ownership
Change Management Team	Manage emergency changes	Structured change requests	Integrated change control
ServiceNow Administrator	Maintain platform	Scalable and trackable solution	Controlled process management

3.5 Module Integration Design

The project integrates the following ServiceNow capabilities:

- Incident Management
- Knowledge Management
- Agent Assist
- Change Management
- Service Management
- Configuration Management
- SLA Management
- Related Lists
- Child Incidents

4. Project Planning Phase

4.1 Project Execution Roadmap

Project Name: Incident Lifecycle Automation in ServiceNow

The project is planned to automate and standardize the complete incident management lifecycle in ServiceNow. The project activities are divided into milestones covering service configuration, incident creation, classification, knowledge assistance, escalation, change management, resolution, and knowledge creation.

Milestone	Planned Activity
Milestone 1	Create Service
Milestone 2	Create Service Offering
Milestone 3	Create Incident
Milestone 4	Classify Incident
Milestone 5	Integrate Knowledge
Milestone 6	Update Watch List
Milestone 7	Reassign Incident
Milestone 8	Level 2 Investigation
Milestone 9	Create Emergency Change
Milestone 10	Create Child Incident
Milestone 11	Add Cause
Milestone 12	Add Resolution
Milestone 13	Create Knowledge Article
Milestone 14	Final Validation
Milestone 15	Testing and Deployment

4.2 Task Planning

Phase A – Service Configuration

Objective: Configure the required service and service offering for the incident.

1. Create **Remote Access** service.
2. Set the operational status.
3. Create **Corporate VPN** service offering.
4. Associate the offering with **Remote Access**.

Deliverable: Remote Access service with Corporate VPN service offering.

Phase B – Incident Management

Objective: Create and classify a VPN-related incident.

1. Impersonate **Service Desk** user.
2. Create a **VPN incident**.
3. Add caller and short description.
4. Assign the incident to **Service Desk**.
5. Update incident details.
6. Select **Network** → **VPN** category.
7. Set urgency and priority.
8. Associate service, service offering, and CI.

Deliverable: Properly created and classified VPN incident.

Phase C – Knowledge Management

Objective: Use existing knowledge to assist in resolving the incident.

1. Open **Agent Assist**.
2. Search for relevant VPN Knowledge Articles.
3. Open the appropriate article.
4. Mark the article as **Helpful**.
5. Attach the article to the incident.
6. Add relevant comments.

Deliverable: Relevant Knowledge Article attached to the incident.

Phase D – Collaboration and Escalation

Objective: Collaborate with stakeholders and escalate the incident to the appropriate support team.

1. Add stakeholder to the **Watch List**.
2. Add user to the **Work Notes List**.
3. Reassign the incident to **Network Support**.
4. Verify the **Assigned To** field.
5. Impersonate **Level 2 Technician**.
6. Assign the incident to the technician.
7. Verify incident and SLA information.

Deliverable: Incident successfully escalated to Level 2 support.

Phase E – Change and Child Incident Management

Objective: Manage additional work required to resolve the incident.

1. Update the **Configuration Item**.
2. Set the incident to **On Hold**.

3. Select **Awaiting Change**.
4. Create and link an **Emergency Change**.
5. Create a **Child Incident**.
6. Verify the parent-child relationship.

Deliverable: Emergency change and child incident linked to the parent incident.

Phase F – Resolution

Objective: Document the cause and resolution and complete the incident.

1. Add the **probable cause**.
2. Add the **resolution code**.
3. Add **resolution notes**.
4. Resolve the parent incident.
5. Verify the child incident status.
6. Verify the change request relationship.

Deliverable: Incident resolved with complete resolution information.

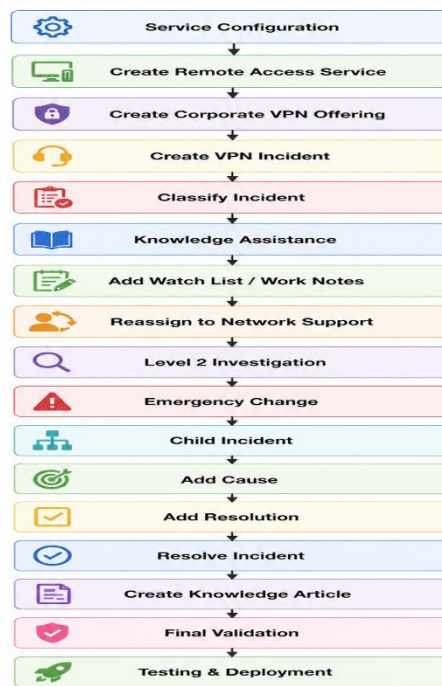
Phase G – Knowledge Creation

Objective: Create reusable knowledge based on the incident resolution.

1. Select **Create Knowledge**.
2. Select **IT Knowledge Base**.
3. Select **Standard** template.
4. Verify the article information.
5. Save the Knowledge Article.
6. Verify the article under **Related Records**.

Deliverable: New Knowledge Article created from the incident resolution.

4.3 Project Workflow



4.4 Project Deliverables

Phase	Deliverable
Service Configuration	Remote Access Service and Corporate VPN Offering
Incident Management	Created and classified VPN Incident
Knowledge Management	Knowledge Article attached to Incident
Collaboration & Escalation	Incident assigned to Level 2 Network Support
Change Management	Emergency Change linked to Incident
Child Incident Management	Parent-Child Incident relationship
Resolution	Cause and Resolution details
Knowledge Creation	New IT Knowledge Article
Final Phase	Validated and tested project

4.5 Expected Project Outcome

At the end of the project, the Incident Lifecycle Automation in ServiceNow will demonstrate the complete incident management process from service configuration and incident creation to classification, knowledge assistance, escalation, change management, resolution, and knowledge creation. This ensures that incidents are handled in a structured and traceable manner.

5. Project Development Phase

This phase contains the actual implementation activities performed in ServiceNow.

5.1 Service Configuration

Step 1: Create Service

Navigate to:

All → cmdb_ci_service.list → New

Enter:

Field	Value
Name	Remote Access
Operational Status	Operational

Submit the record.

The screenshot shows the ServiceNow 'New record' form for 'Service'. The form is titled 'Service New record' and has a 'Submit' button in the top right corner. The form contains several fields:

- Name: Remote Access
- Model ID: (empty)
- Owned by: (empty)
- Business criticality: 2 - somewhat critical
- Version: (empty)
- Used for: Production
- Operational status: Operational
- Service classification: Business Service
- Comments: (empty text area)
- Approval group: (empty)
- Support group: (empty)
- Change Group: (empty)
- Managed by: (empty)
- SLA: (empty)
- Location: (empty)
- Vendor: (empty)

At the bottom of the form, there is a 'Submit' button and a section titled 'Related Links' with two links: 'Convert to business.service' and 'Convert to technology.management.service'.

Step 2: Create Service Offering

Navigate to:

All → service_offering.list → New

Enter:

Field	Value
Name	Corporate VPN
Parent	Remote Access
Operational Status	Operational

Submit the record.

Offering New record

Name: Corporate VPN

Model ID:

Owned by:

* Parent: Remote Access

Used for: Production

Operational status: Operational

Vendor:

Approval group:

Support group:

Change group:

Managed by:

Version:

Business criticality: 2 - somewhat critical

SLA:

Location:

Short Description:

Description:

Comments:

Contract:

Submit

5.2 Incident Creation

Navigate to:

Service Operations Workspace → List → Incidents → Open → New

Create an incident with:

Field	Value
Short Description	Unable to connect to Corporate VPN from home office
Caller	Michael Hoefer
Assignment Group	Service Desk

Save the incident.

servicenow All Favorites History Workspaces Service Operations Workspace Search

Create New Incident Save Assign to me

Incident

Short description *
Unable to connect to Corporate VPN from home office

Description

Number
INC0010001

State
New

Caller *
Michael Hoefer

Location
SHS quadra 5, Bloco E., Brasilia

Channel
-- None --

Impact
3 - Low

Urgency
3 - Low

Priority
4 - Low

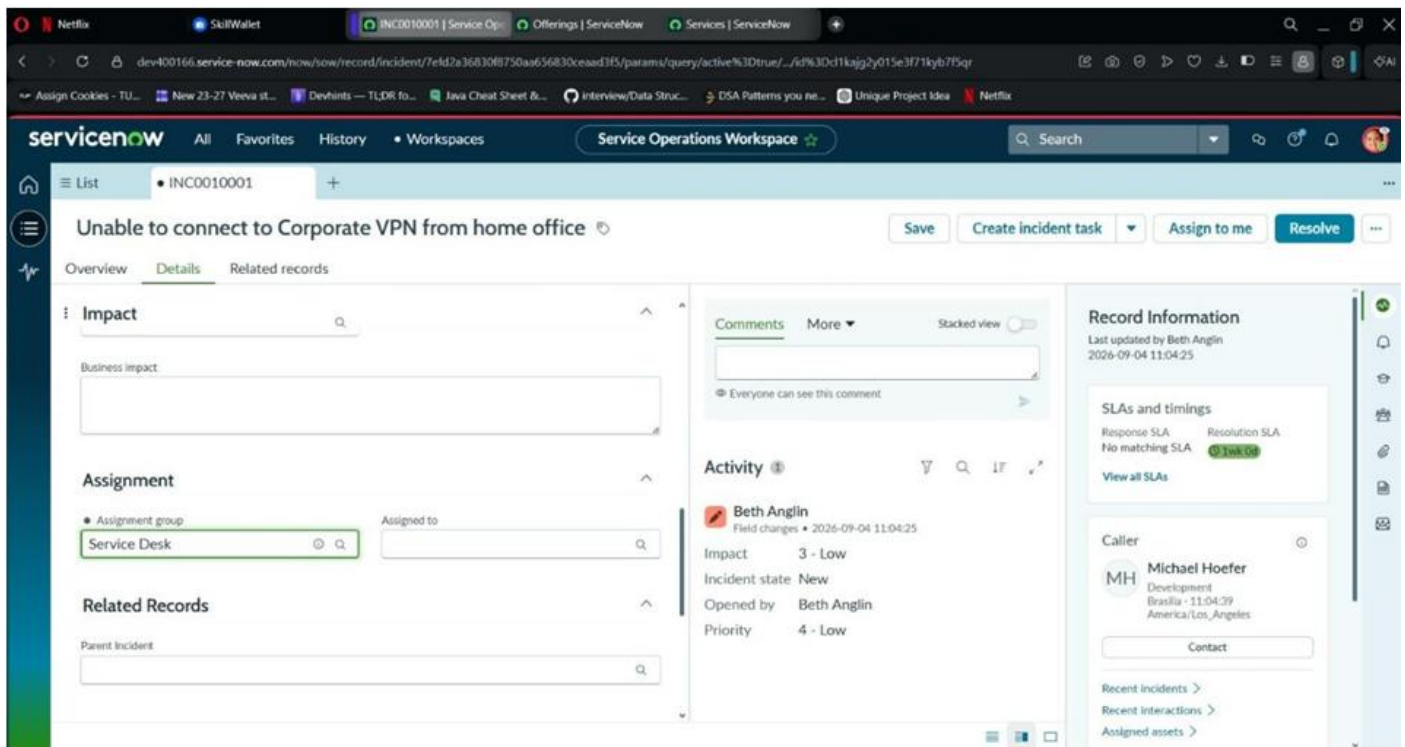
Agent Assist

Search: Unable to connect to Corp...

Article: How to configure VPN for Apple ...
How to configure VPN for Apple Devices For an iPhone or iPad running iOS? Select Sett...
updated 12 years ago · 2 views

Article: Automatic Replies (Out Of Office)
Automatic Replies (Out Of Office) This article explains how to use automatic replies in...
updated 4 years ago · 85 views

Article: Excel Functionality
Excel Functionality Copying formulas without changing cell locations When you want to...



5.3 Incident Classification

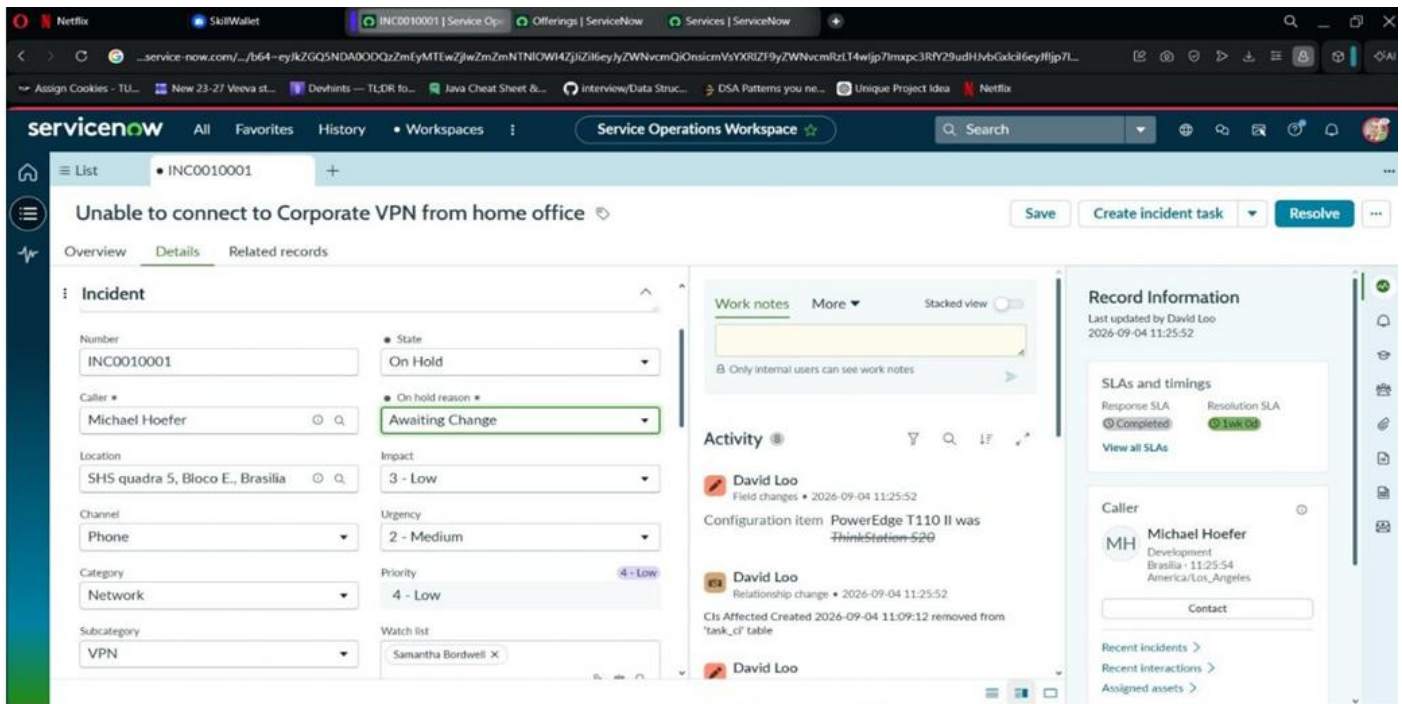
Update the incident with:

Field	Value
Description	Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine.
Channel	Phone
Category	Network
Subcategory	VPN
Urgency	2 – Medium
Service	Remote Access
Service Offering	Corporate VPN
Configuration Item	ThinkStationS20

Add the following work note:

Classified and assigned incident.

Then select **Assign to me** and save.



5.4 Agent Assist Implementation

Open **Agent Assist** from the right-side panel.

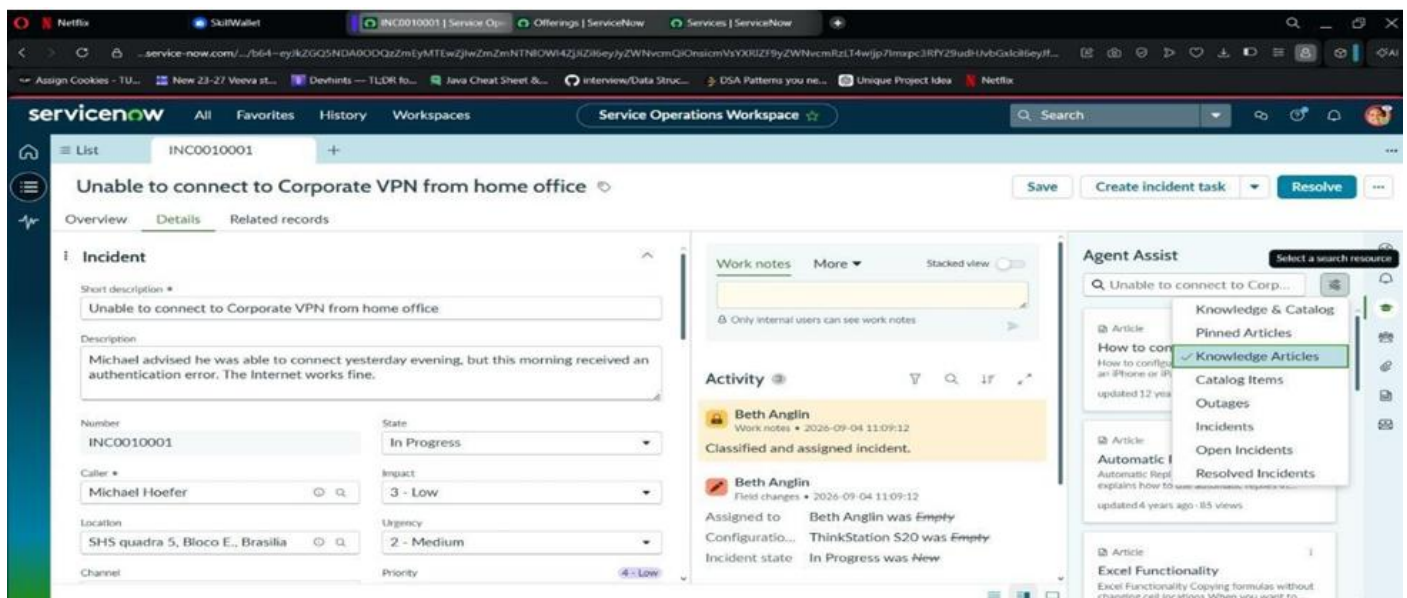
Verify that relevant Knowledge Articles are displayed based on the incident's short description.

Perform the following:

1. Open a relevant Knowledge Article.
2. Select the three-dot menu.
3. Mark the article as **Helpful**.
4. Select **Attach Article**.
5. Add the comment:

Follow the steps and resolve the incident.

6. Attach the article.
7. Return to the incident.
8. Verify that the article is recorded in the Activity log.



5.5 Watch List Configuration

In the incident **Details** tab:

- Add **Samantha Bordwell** to Watch List.
- Add **Beth Anglin** to Work Notes List.

This ensures that relevant stakeholders receive updates and can monitor the incident.

The screenshot shows the ServiceNow interface for an incident titled "Unable to connect to Corporate VPN from home office" (ID: INC0010001). The "Details" tab is active. The incident is categorized as "SHS quadra 5, Bloco E., Brasilia" with a "2 - Medium" priority. The channel is "Phone" and the category is "Network". The subcategory is "VPN". The watch list includes "Samantha Bordwell" and the work notes list includes "Beth Anglin".

Field	Value
Incident	SHS quadra 5, Bloco E., Brasilia
Channel	Phone
Category	Network
Subcategory	VPN
Priority	2 - Medium
Watch list	Samantha Bordwell
Work notes list	Beth Anglin

5.6 Incident Reassignment

Update:

Assignment Group → **Network**

Verify that the **Assigned To** field is cleared automatically when the assignment group changes.

Save the incident.

The screenshot shows the ServiceNow interface for the same incident. The "Impact" field is empty. The "Assignment" section shows the "Assignment group" set to "Network" and the "Assigned to" field is empty.

Field	Value
Impact	
Assignment group	Network
Assigned to	

5.7 Level 2 Investigation

Impersonate the Level 2 Network technician.

Navigate to:

Service Operations Workspace → Incidents → Open

Open the VPN incident.

Select **Assign to me**.

Verify:

- Incident description is available.
- Knowledge Articles are available.
- SLA information is visible.
- Task SLAs appear in Related Records.
- Incident information is accessible to the Network team.

The screenshot displays the ServiceNow interface for an incident record. The browser address bar shows the URL: `dev400166.service-now.com/now/sow/record/incident/7efd2a36830f8750aa656830cead3f5`. The page title is "Unable to connect to Corporate VPN from home office". The incident number is INCO010001. The status is "In Progress". The priority is "3 - Low". The impact is "3 - Low". The description states: "Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine." The "Compose" section shows "Work notes" and "Activity". The "Record Information" section shows "SLAs and timings" with "Response SLA" as "Completed" and "Resolution SLA" as "1 wk Up". The "Caller" section shows "Michael Hoefer" with contact information. The "Recent incidents", "Recent interactions", and "Assigned assets" sections are also visible.

5.8 Emergency Change Integration

Update the Configuration Item as required.

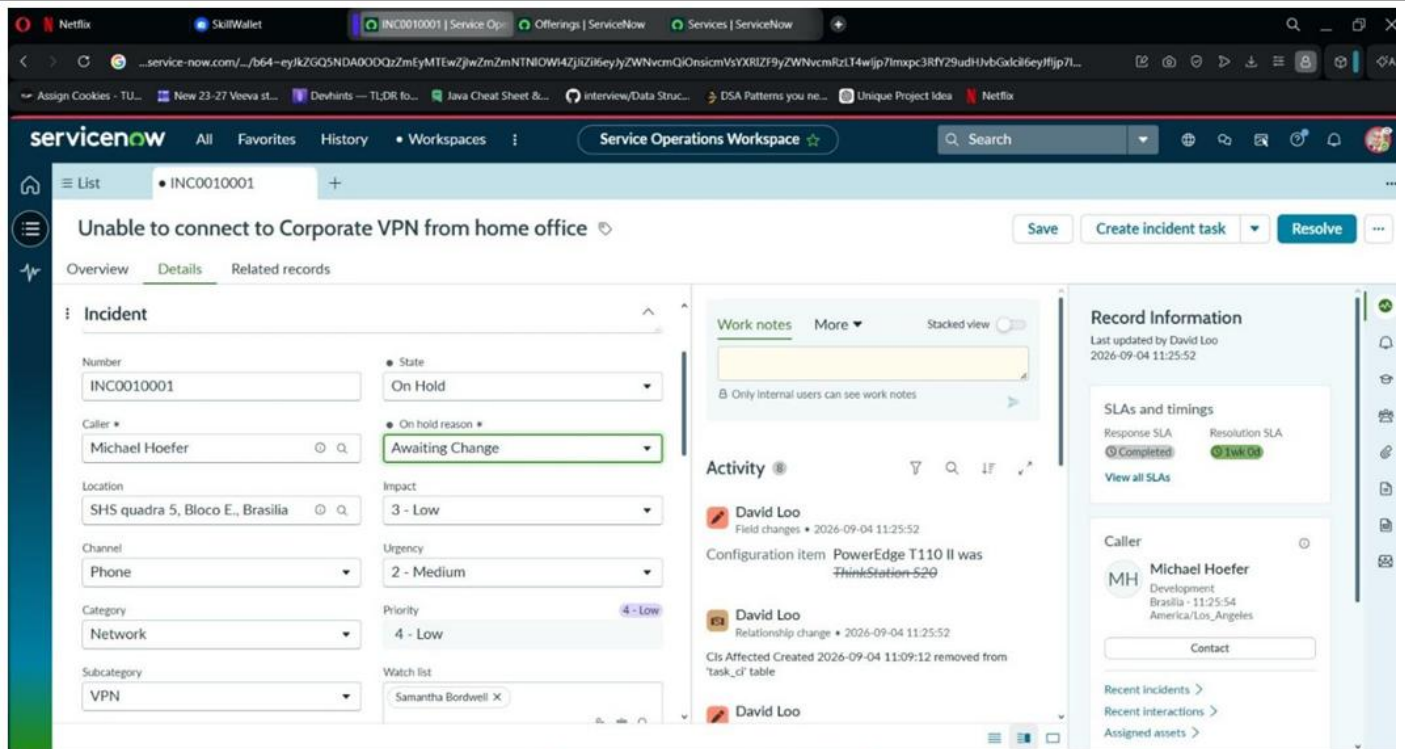
Place the incident in:

State: On Hold

On Hold Reason: Awaiting Change

Create or associate the required emergency change request with the incident.

The change request provides controlled implementation of the technical fix while maintaining traceability between the incident and change.



5.9 Child Incident Creation

Navigate to:

Related Records → Child Incidents → New

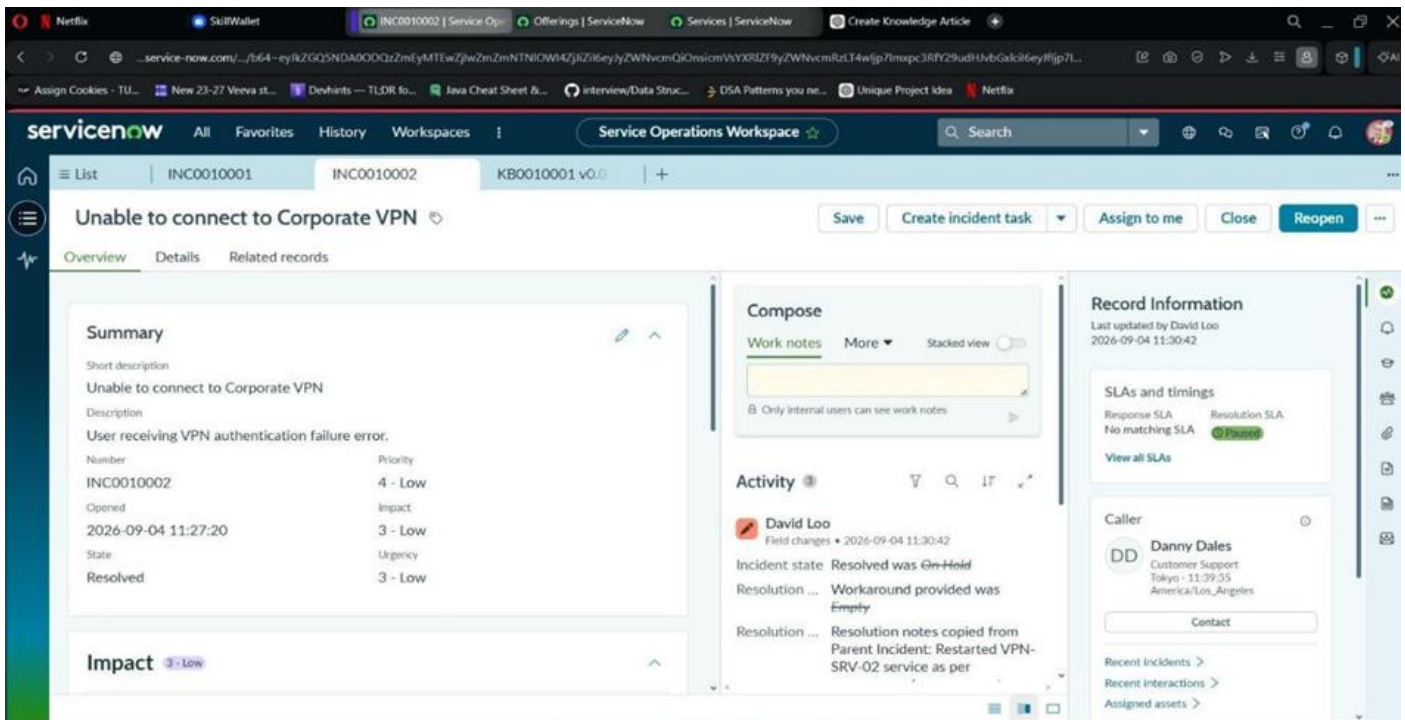
Enter:

Field	Value
Short Description	Unable to connect to Corporate VPN
Caller	Any available user
Description	User receiving VPN authentication failure error

Save the record.

Record the child incident number.

Return to the parent incident and verify that the child incident appears under **Related Records**.



5.10 Cause Documentation

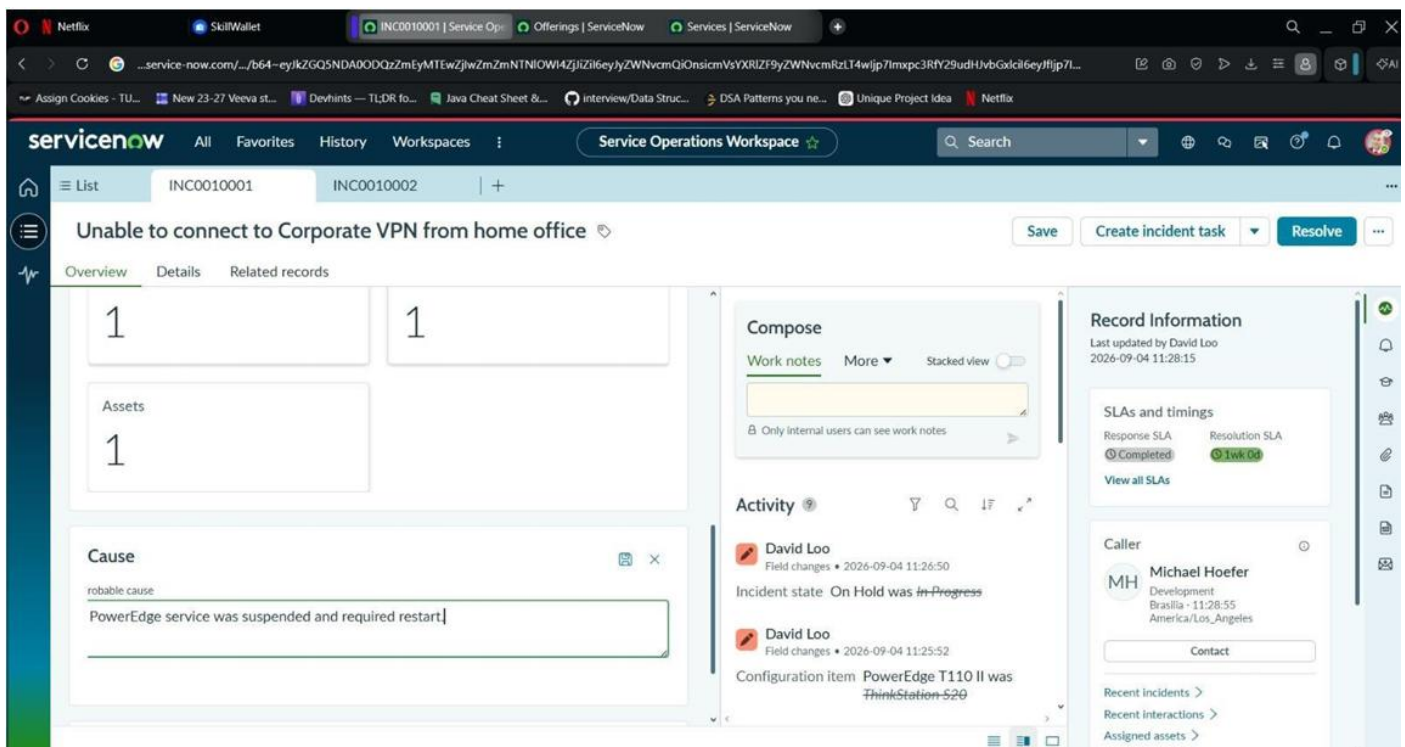
From the Overview tab, select **Add Cause**.

Enter:

Probable Cause:

PowerEdge service was suspended and required restart.

Save the cause information.



5.11 Resolution Documentation

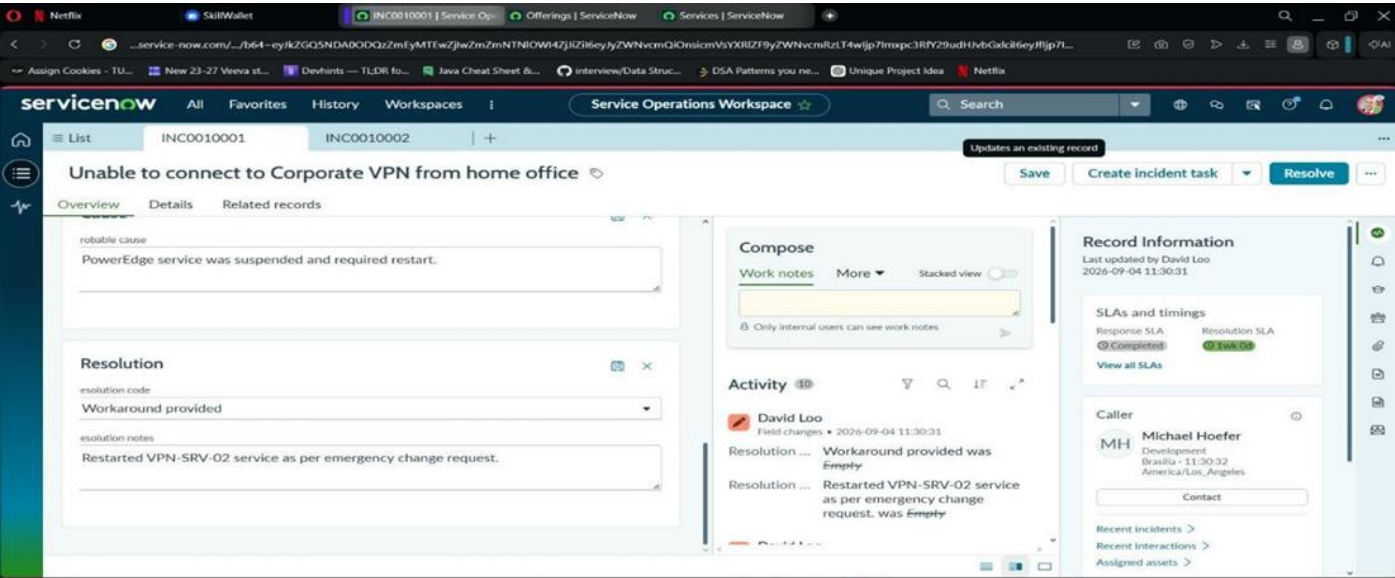
Select **Add Resolution**.

Enter:

Field	Value
Resolution Code	Workaround provided
Resolution Notes	Restarted VPN-SRV-02 service as per emergency change request.

Save the resolution.

Select **Resolve** and confirm the resolution.



5.12 Knowledge Article Creation

After resolving the incident, select **Create Knowledge**.

Configure:

Field	Value
Knowledge Base	IT
Article Template	Standard

Select **Next**.

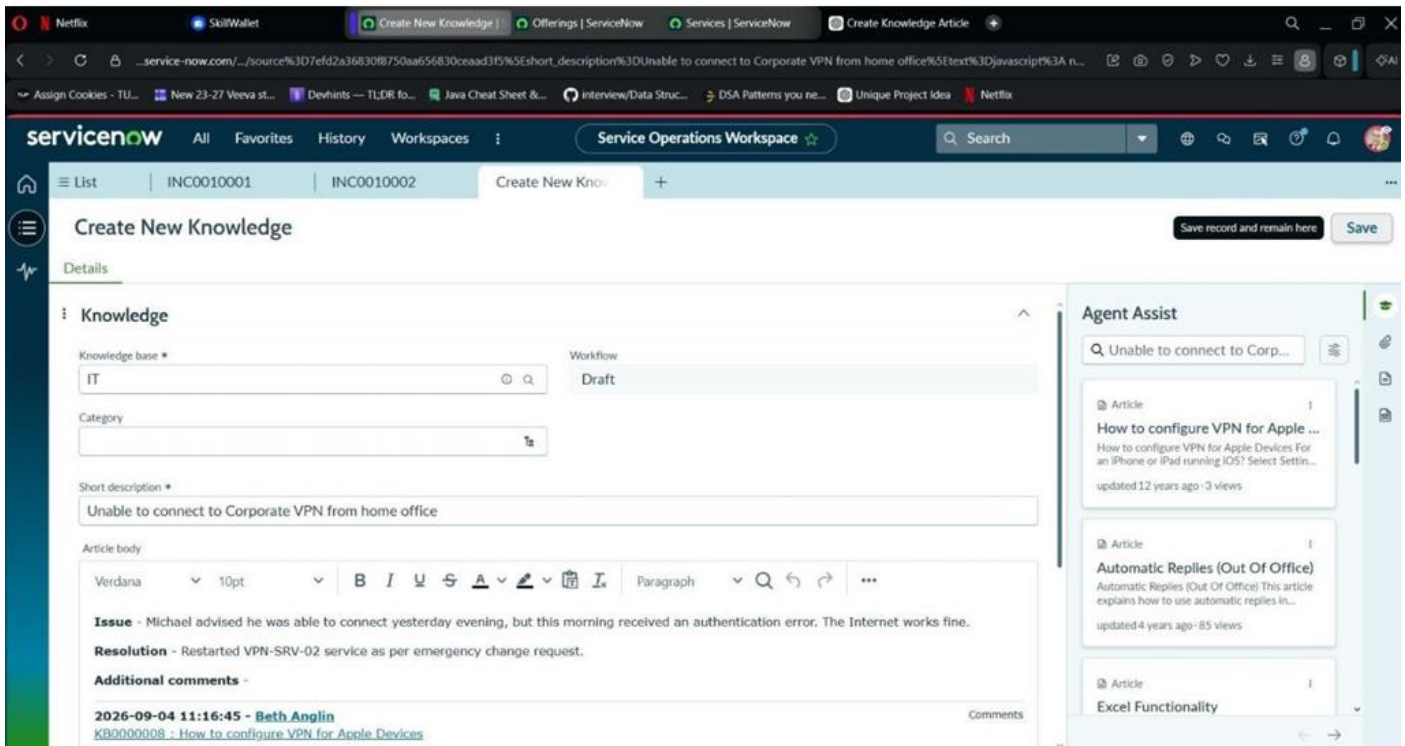
Verify the incident and resolution details.

Save the Knowledge Article.

Return to the parent incident and navigate to:

Related Records → **Knowledge**

Verify that the newly created Knowledge Article is linked to the incident.



5.13 Testing and Validation

The implemented solution should be tested against the complete incident lifecycle.

Test Case 1 – Incident Creation

Expected Result:

Incident should be successfully created with caller and short description.

Test Case 2 – Classification

Expected Result:

Category, subcategory, urgency, service, service offering, and CI should be correctly populated.

Test Case 3 – Knowledge Integration

Expected Result:

Relevant Knowledge Articles should appear in Agent Assist and should be attachable to the incident.

Test Case 4 – Reassignment

Expected Result:

Incident should move from Service Desk to Network support.

Test Case 5 – SLA

Expected Result:

Applicable SLA should be visible under Task SLAs.

Test Case 6 – Change Integration

Expected Result:

Emergency Change Request should be linked to the incident.

Test Case 7 – Child Incident

Expected Result:

Child incident should be created and displayed under the parent incident.

Test Case 8 – Resolution

Expected Result:

Cause and resolution information should be recorded and the incident should move to Resolved.

Test Case 9 – Knowledge Creation

Expected Result:

A Knowledge Article should be created from the resolved incident and linked to it.

5.14 Final Validation Checklist

At the end of development and testing, verify:

- ☒ Parent incident is **Resolved**.
- ☒ Child incident is **Resolved**.
- ☒ Incident activity contains the resolution information.
- ☒ Change Request is linked.
- ☒ Knowledge Article is created.
- ☒ Knowledge Article is linked to the incident.
- ☒ SLA tracking is visible.
- ☒ Assignment and reassignment work correctly.
- ☒ Incident classification is accurate.
- ☒ Watch List and Work Notes List are populated.
- ☒ Parent-child incident relationship is maintained.
- ☒ Complete incident history is available.